

STA201: Elements of Statistics and Probability

Lecture 3 & 4

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1. Measures of central tendency
 - Mean
 - Median
 - Mode
2. Calculation of the measures using raw (ungrouped) and grouped data.
3. Weighted mean, Geometric mean and Harmonic mean.
4. Median like measures Quartiles, Deciles, Percentiles.

Measures of Central tendency

- The measure of central tendency is the statistical measure that identifies a single value as representative of an entire distribution.
- It is also known as the measure of location or the measure of average or simply the averages.
- Simpson and Kafka defined it as A measure of central tendency is a typical value around which other figures congregate.
- In most datasets i.e. population or sample, the values show a distinct tendency to group or cluster around a central point.
- This tendency of clustering the values around the center of the series is usually called central tendency.

Why central tendency is so important?

- It aims to provide an accurate description of the entire data.
- It is the single value that is most typical/ representative of the collected data.
- Central tendency can be considered as the first summary of the data.

Measures of central tendency

There are different measures of central tendency.

- Mean
 - Arithmetic mean
 - Geometric mean
 - Harmonic mean
- Median
- Mode

Arithmetic Mean

The Arithmetic Mean

- Arithmetic mean is a mathematical average and it is the most popular measures of central tendency.

$$\text{Arithmetic Mean (AM)} = \frac{\text{Sum of all the values}}{\text{Total number of data points}}$$

- Arithmetic mean (AM) for
 - Sample is denoted by $\bar{x}$
 - Population is denoted by μ
- For a set of observations $x_1, x_2, \dots, x_n$, the following is the formula for the Arithmetic mean

$$\begin{aligned}\bar{x} &= \frac{1}{n} \sum_{i=1}^n x_i \\ &= \frac{1}{n} (x_1 + x_2 + \dots + x_n)\end{aligned}$$

Example: For Raw Data

Example

- The marks (out of 25) obtained by 8 students in a class test are 10, 19, 12, 21, 18, 20, 11, and 19. What is the arithmetic mean of the marks obtained by the students?

Solution

- To find Arithmetic mean of the marks obtained by the students Using the arithmetic mean formula,

$$\begin{aligned}\text{Arithmetic mean} &= \frac{1}{n} \sum_{i=1}^n x_i = \frac{\text{Sum of observations}}{\text{Number of observations}} \\ &= \frac{(10 + 19 + 12 + 21 + 18 + 20 + 11 + 19)}{8} \\ &= 16.25\end{aligned}$$

Exercise

1. There are 10 salespeople employed by Midtown Ford. The number of new cars sold last month by the respective salespeople were: 15, 23, 4, 19, 18, 10, 10, 8, 28, 19. Find the average number of cars sold last month.
2. A mail-order company counted the number of incoming calls per day to the companys toll-free number during the first 7 days in May: 14, 24, 19, 31, 36, 26, 17. Find the average number of incoming calls per day.
3. AAA Heating and Air Conditioning completed 30 jobs last month with a mean revenue of \$5,430 per job. The president wants to know the total revenue for the month. Based on the limited information, can you compute the total revenue? What is it?

Arithmetic Mean for Grouped Data

Recall the notational form of a frequency distribution table for grouped data:

Class Interval	Class mid value	Frequency
$L_1 - U_1$	x_1	f_1
$L_2 - U_2$	x_2	f_2
$L_3 - U_3$	x_3	f_3
$\dots$	$\dots$	$\dots$
$L_i - U_i$	x_i	f_i
$\dots$	$\dots$	$\dots$

Where, f_i and x_i denote the class frequency and class mid value of the i^{th} class ($i = 1, 2, \dots, k$) respectively. The mean can be obtained as:

$$\bar{x} = \frac{\sum_{i=1}^k f_i x_i}{\sum_{i=1}^k f_i}$$

Example: Arithmetic Mean for Group data

Table 1: Distribution of Height (in inches) of 100 University students

Height (inch)	Frequency (f_i)	Mid values (x_i)	$f_i x_i$
60-63	5	61.5	307.5
63-66	18	64.5	1161
66-69	42	67.5	2835
69-72	27	70.5	1903.5
72-75	8	73.5	588
Total	100		6795

$$\begin{aligned}\bar{x} &= \frac{\sum_{i=1}^n f_i x_i}{\sum_{i=1}^n f_i} \\ &= \frac{6795}{100} \\ &= 67.95\end{aligned}$$

Properties of Arithmetic Mean

- Every set of interval and ratio level data has a mean.
- A set of data has only one mean. The mean is unique.
- All the values are included in computing the mean.
- The mean is a useful measure for comparing two or more populations.
- The arithmetic mean is the only measure of central tendency where the sum of the deviations of each value from the mean will always be zero. Expressed symbolically:

$$\sum_{i=1}^n (x_i - \bar{x}) = 0$$

Advantages and Disadvantages of Arithmetic mean

Advantages

- It is well defined, easy to understand & simple to calculate.
- It includes all the observations.
- It is not based on the position in the series.
- It provides equal weight to all the observations.

Disadvantages

- It is affected by extreme values.
- It cannot be calculated for open ended classes.
- Cannot be used in case of qualitative data.
- It can not be located graphically.

Median

Median for Raw Data

- Median is a central value of the distribution.
- It divides the distribution in two equal parts, each part containing equal number of items.
- Thus it is the central value of the variable, when the values are arranged in order of magnitude.
- Median is usually denoted by M_e .

How to calculate Median

For a set of observations $x_1, x_2, \dots, x_n$, the following is the formula for the median

1. When n is odd

- order the observations
- $M_e = \frac{(n+1)}{2}^{\text{th}}$ ordered observation

2. When n is even

- order the observations
- $M_e = \frac{1}{2} \{ (\frac{n}{2})^{\text{th}} + (\frac{n}{2} + 1)^{\text{th}} \}$ ordered observation

Example: n is Odd

- The ages of a family of seven members are given as 12, 7, 2, 34, 17, 21 and 19. Find the median age.

Formula

- Step 1: Arrange the values in ascending order : 2, 7, 12, 17, 19, 21, 34
- Step 2: Count the total number of elements, $n = ?$ Here $n = 7$, an odd number.
- Step 3:

$$\begin{aligned}M_e &= \text{Value of } \frac{(n+1)^{th}}{2} \text{ observation} \\&= \text{Value of } \frac{(7+1)^{th}}{2} \text{ observation} \\&= \text{Value of } 4^{th} \text{ observation} = 17\end{aligned}$$

- Step 4: Median Age of the family is 17 years.

Example: n is Even

- The ages of a family of eight members are given as 12, 7, 2, 34, 17, 21, 19 and 40. Find the median age.

Formula

- Step 1: Arrange the values in ascending order : 2, 7, 12, 17, 19, 21, 34, 40
- Step 2: Count the total number of elements, $n = ?$ Here $n = 8$, an even number.
- Step 3:

$$\begin{aligned}M_e &= \frac{1}{2} \left\{ \left(\frac{n}{2} \right)^{\text{th}} \text{observation} + \left(\frac{n}{2} + 1 \right)^{\text{th}} \text{observation} \right\} \\&= \frac{1}{2} \{ 4^{\text{th}} \text{observation} + 5^{\text{th}} \text{observation} \} \\&= \frac{1}{2} (17 + 19) = 18\end{aligned}$$

- Step 4: Median Age of the family is 18 years.

Test Yourself

The following data represents the amount (in thousands taka) of loan requirements of the people of two different upazilla. Using median, comment on which upazilla has the greater average demand of loans:

Upazilla 1	42	12	26	18	9	35	28	39	8
Upazilla 2	8	15	10	18	22	20	26	42	35

Median for Grouped Data

Median

$$M_e = L_0 + \frac{\frac{n}{2} - F_{-1}}{f_m} \times C$$

Where,

- L_0 = Lower limit of the median class,
- n = Number of items in the data,
- F_{-1} = Cumulative frequency of the pre median class,
- f_m = Frequency of the median class,
- C = Class interval of median class.

Median Class: the class that contains $\frac{n}{2}$ th observation of the given data

Median: Example for Group Data

Height (inch)	Frequency (f_i)	Cumulative Frequency
60-63	5	5
63-66	18	23
66-69	42	65
69-72	27	92
72-75	8	100

- $\frac{n}{2} = \frac{100}{2} = 50^{\text{th}}$ ordered observation, i.e. the class of 66-69.
- L_0 = Lower class boundary of the median class = 66,
 n = Number of items in the data = 100,
 F_{-1} = Cumulative frequency of the pre median class = 23,
 f_m = Frequency of the median class = 42,
 C = Class interval of median class = 3.

$$\text{Median, Me} = L_0 + \frac{\frac{n}{2} - F_{-1}}{f_m} \times C = 66 + \frac{\frac{100}{2} - 23}{42} \times 3 = 67.93$$

Median Calculation from Ogive Curve

How to calculate median from Ogive or cumulative frequency curve.

Follow the class Lecture.

Properties of Median

- Every set of ordinal, interval and ratio level data has a median.
- The median is unique; that is, like the mean, there is only one median for a set of data.
- It is not affected by extremely large or small values and is therefore a valuable measure of central tendency when such values do occur.
- It can be computed for a frequency distribution with an open-ended class if the median does not lie in an open ended class.

Advantages and Disadvantages of Median

Advantages

- Median can be calculated in all distributions.
- Median can be ascertained even with the extreme items.
- It can be located graphically
- It is most useful in dealing with qualitative data.

Disadvantages

- It is not based on all the values.
- It is not capable of further mathematical treatment.
- It is affected by fluctuation of sampling.
- In case of even number of values it may not a value from the data.

Usage of Median

- Whenever a data set has extreme values, the median is the preferred measure of central location.
- The median is the measure of location most often reported for annual income and property value data.
- A few extremely large incomes or property values can inflate the mean.

Mode

- Mode is the most frequent value or score in the distribution.
- Generally speaking, mode can be used to describe qualitative data.
- Mode is particularly useful average for discrete data.
- For ungrouped data / categorical variable: mode is the value of the variable for which the frequency is highest

Mode for Raw Data

Example 1

- Find the Mode of the data set 3, 7, 5, 13, 20, 23, 39, 23, 40, 23, 14, 12, 56, 23, 29.
- Count how many times each of the number occurs (can be counted easily by writing them in order): 3, 5, 7, 12, 13, 14, 20, 23, 23, 23, 23, 29, 39, 40, 56
- In this case the mode is 23.

Example 2

- 1, 3, 3, 3, 4, 4, 6, 6, 6, 9 .
- 3 appears three times, as does 6. So there are two modes: 3 and 6.

Example 3

- 4, 7, 11, 16, 20, 22, 25, 26, 33.
- There exists no Mode.

Example 4

- strongly disagree, disagree, somewhat disagree, neutral, somewhat agree, agree, strongly agree, strongly agree, strongly agree.
- "strongly agree" appears 3 times, so the mode is "strongly agree".

Example 5

- Ten customers answered the following to the question "Which of the following items do you want for your pizza toppings?"
- Data: Spinach, Pepperoni, Sardines, Olives, Sardines, Sausage, Extra cheese, Sardines, Onions and Tomatoes.
- Count how many times each of the value occurs Spinach, Pepperoni, **Sardines**, Olives, **Sardines**, Sausage, Extra cheese, **Sardines**, Onions and Tomatoes.
- Sardines appears 3 times, so the Mode is "**Sardines**".

Determination of Mode: Grouped Data

Mode

$$M_o = L_0 + \frac{(f_0 - f_{-1})}{(f_0 - f_{-1}) + (f_0 - f_1)} \times C$$

Where,

- L_0 = Lower limit of the modal class,
- f_0 = Frequency of the modal class,
- f_{-1} = Frequency of the pre modal class,
- f_1 = Frequency of the post modal class,
- C = Class interval of the modal class

Modal class: The class with the largest frequency.

Mode for Grouped data

Height (inch)	Frequency (f_i)
60-63	5
63-66	18
66-69	42
69-72	27
72-75	8

- f_0 = Frequency of the modal class = 42,
- f_{-1} = Frequency of the pre modal class = 18,
- f_1 = Frequency of the post modal class = 27,
- L_0 = Lower limit of the modal class = 66 ,
- C = Class interval of the modal class = 3

$$\text{Mode, } M_o = L_0 + \frac{(f_0 - f_{-1})}{(f_0 - f_{-1}) + (f_0 - f_1)} \times C = 66 + \frac{24}{24 + 15} \times 3 = 67.84.$$

Properties of Mode

- Unlike the mean and median, **Mode is not unique**; that is, there may be more than one mode for a set of data.
 - If the data have exactly **two modes**, the data are **bimodal**.
 - If the data have **more than two modes**, the data are **multimodal**.
- Mode may be calculated for **nominal level, ordinal level, interval level and ratio level data**.
- Mode of data set may not exist.

Advantages and Disadvantages of Mode

Advantages

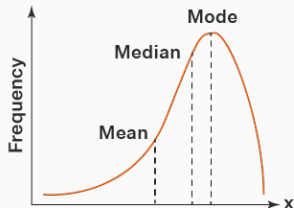
- Mode is readily comprehensible and easily calculated
- It is the best representative of data
- It is not at all affected by extreme value.
- The value of mode can also be determined graphically.

Disadvantages

- It is not based on all observations.
- It is not capable of further mathematical manipulation.
- Mode is affected to a great extent by sampling fluctuations.
- Choice of grouping has great influence on the value of mode.

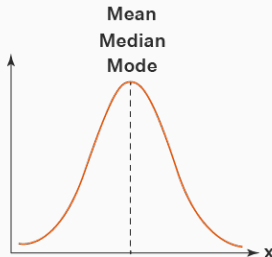
Relationship of Mean, Median and Mode

mean < median < mode



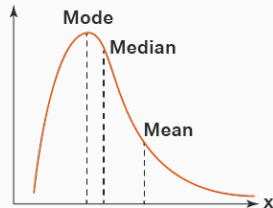
Negatively Skewed

mean = median = mode



Symmetrical Distribution

mean > median > mode



Positively Skewed

Weighted Mean

Weighted Mean

- The weighted mean is **a special case of the arithmetic mean.**
- It occurs when there are several observations of the same value.
- Instead of each data point contributing equally to the final mean, some data points contribute more weight than others.
- **Note: If all the weights are equal, then the weighted mean equals the arithmetic mean.**

Weighted Mean: Example

- Suppose, Shumi's Hot Cake offers three different kinds of burger packages small, medium and large for Tk. 100, Tk. 125 and Tk. 150 respectively.
- Of the last 10 burgers sold 3 were small, 4 were medium and 3 were large
- Using above information, the mean selling price of the last 10 burger packages sold:

$$\begin{aligned}\bar{x}_w &= \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i} \\ &= \frac{(3 * 100) + (4 * 125) + (3 * 150)}{(3 + 4 + 3)} \\ &= \frac{1250}{10} = 125\end{aligned}$$

Weighted Mean: Another Example

- A student may use a weighted mean in order to calculate his/her percentage grade in a course. In such an example, the student would multiply the weighing of all assessment items in the course (e.g., assignments, exams, quizzes, etc.) by the respective grade that was obtained in each of the categories.
- Consider a student with the following grades:

Item	Weight	Grade
Attendance	5%	70%
Assignment	15%	80%
Quiz	20%	85%
Midterm Exam	25%	70%
Final Exam	35%	80%

Weighted Mean: Example

$$\begin{aligned}\sum w_i x_i &= (5 \times 0.7) + (15 \times 0.8) + (20 \times 0.85) + (25 \times 0.7) + (35 \times 0.8) \\ &= 78\end{aligned}$$

$$\sum w_i = 100$$

$$\text{Final grade} = \frac{78}{100} = 78\%$$

- Here, we see that the student is actually able to get a better than expected grade by doing well in the most heavily weighted component of the course: the final exam.
- Given the knowledge of the weighing of each assessment element in the course, students can allocate their study time more effectively.

Exercise

1. The Carter Construction Company pays its hourly employees \$16.50, \$19.00, or \$25.00 per hour. There are 26 hourly employees, 14 of whom are paid at the \$16.50 rate, 10 at the \$19.00 rate, and 2 at the \$25.00 rate. What is the mean hourly rate paid the 26 employees? (**Ans: 18.12\$**)
2. Andrews and Associates specialize in corporate law. They charge \$100 an hour for researching a case, \$75 an hour for consultations, and \$200 an hour for writing a brief. Last week one of the associates spent 10 hours consulting with her client, 10 hours researching the case, and 20 hours writing the brief. What was the mean hourly charge for her legal services? (**Ans: 143.75\$**)
3. The Bookstall Inc. is a specialty bookstore concentrating on used books sold via the Internet. Paperbacks are \$1.00 each, and hardcover books are \$3.50. Of the 50 books sold last Tuesday morning, 40 were paperback and the rest were hard- cover. What was the mean price of a book? (**Ans: 1.5\$**)

Geometric mean

The Geometric mean

- Geometric mean is often used for a set of numbers whose values are meant to be multiplied together or are exponential in nature.
- For example, data related to **growth rates** (i.e. population growth) or **interest rates** of a financial investment.
- We conduct arithmetic mean when observation are independent to each other and perform geometric mean when observations are dependent to each other.

The Geometric mean

For a set of observations $x_1, x_2, \dots, x_n$; ($x_i > 0, \forall i = 1, 2, \dots, n$), the following is the formula for the Geometric mean

$$GM = \left(\prod_{i=1}^n x_i \right)^{\frac{1}{n}} = (x_1 * x_2 * x_3 * \dots * x_n)^{\frac{1}{n}}$$

$$\Rightarrow \log GM = \log (x_1 * x_2 * x_3 * \dots * x_n)^{\frac{1}{n}}$$

$$\Rightarrow \log GM = \frac{1}{n} (\log x_1 + \log x_2 + \dots + \log x_n)$$

$$\Rightarrow GM = \text{Antilog} \left\{ \frac{1}{n} \sum_{i=1}^n \log x_i \right\}$$

Example: Find the geometric mean for the values 47, 23, 71 and 81.

$$G.M = (47 \times 23 \times 71 \times 81)^{\frac{1}{4}} = (6216831)^{\frac{1}{4}} = 49.93$$

When to use Geometric mean

- **Geometric mean is most notably used in financial reporting** (for evaluating investment returns, fluctuating interest rates and inflation rates).
- Calculate the average growth rate or compound annual growth rate.
- Handle skewed distributions or data with outliers more effectively than the arithmetic mean.
- **The geometric mean is not suitable for data that includes zero or negative values** since these would result in undefined or complex values.
- **The geometric mean of a data set with at least one zero is always zero, as a result, the geometric mean does not capture any information about the non-zero values.**

Example

- Lets say you own a piece of art that increases in value by 50% the first year after you buy it, 20% the second year, and 90% the third year. In such circumstances at the end of
 - first year the value was multiplied by 150% or 1.5
 - the second year the value **at the end of year 1** was multiplied by 120% or 1.2
 - at the end of the third year the value **at the end of year 2** was multiplied by 190% or 1.9
- Art work value year 0: 90,000*tk*.
 - Art work value year 1: $90,000 \times 1.5 = 135,000*tk*$
 - Art work value year 2: $135,000 \times 1.2 = 162,000*tk*$.
 - Art work value year 3: $162,000 \times 1.9 = 307,000*tk*$.

Example: The Geometric mean

- We can also calculate this using geometric mean.
- The geometric mean of the increasing rates

$$(1.5 \times 1.2 \times 1.9)^{(1/3)} \approx 1.51$$

- Here, this 1.51 means if we multiply our initial investment by 1.51 each year, we would get the same amount as if we had multiplied it by 1.5, 1.2, and 1.9
- So, using geometric mean we get
 $90,000 \times 1.51 \times 1.51 \times 1.51 = 307,000tk.$

Exercise

1. The return on investment earned by Atkins Construction Company for four successive years was 30%, 20%, 40%, and 200%. What is the geometric mean rate of return on investment?
2. Sakib bought a share of 100000 tk. for a particular company in 2018. The growth of the share will increase with 10% during 2019. But due to Covid-19 pandemic it decreases with a 5% rate in 2020. In 2021 the growth rate of share increases again with 15%. In 2022, Sakib wants to withdraw the share, how much will he get?

Test Yourself

The Population of Saint Martins Island changed by the following percentages each year for the last decade ("+" means increase, "-" means decrease)

Year	Change
2010	+2%
2011	+3%
2012	-1%
2013	+5%
2014	+7%
2015	+5%
2016	+2%
2017	-1%
2018	+2%
2019	+3%

Question: What is the average population growth per year for the last decade?

Geometric Mean for Grouped Data

- Notational form of a frequency distribution for grouped data:

Class Interval	Class mid value	Frequency
$L_1 - U_1$	x_1	f_1
$L_2 - U_2$	x_2	f_2
$L_3 - U_3$	x_3	f_3
$\dots$	$\dots$	$\dots$
$L_i - U_i$	x_i	f_i
$\dots$	$\dots$	$\dots$

- For a frequency distribution with f_i and x_i denoting the class frequency and class mid value of the i th class ($i = 1, 2, \dots, k$), the geometric mean can be obtained as:

$$GM = \text{Antilog} \left\{ \frac{1}{N} \sum_{i=1}^k f_i * \log x_i \right\}$$

Where, $N = \sum_{i=1}^k f_i$

Example: Geometric Mean for Grouped data

Table 2: Distribution of Height (in inches) of 100 University students

Height (inch)	Frequency (f_i)	Mid values (x_i)	$f_i * \log x_i$
60-63	5	61.5	8.944
63-66	18	64.5	32.572
66-69	42	67.5	76.831
69-72	27	70.5	49.901
72-75	8	73.5	14.930
Total	$N = \sum f_i = 100$		183.178

$$\begin{aligned} GM &= \text{Antilog} \left\{ \frac{1}{N} \sum_{i=1}^k f_i * \log x_i \right\} \\ &= \text{Antilog} \left(\frac{183.178}{100} \right) \\ &= \text{Antilog} (1.831780) = 67.86 \end{aligned}$$

Harmonic Mean

The Harmonic Mean

- The Harmonic Mean (HM) is defined as the reciprocal of the average of the reciprocals of the data values.
- Harmonic mean gives less weightage to the large values and large weightage to the small values to balance the values correctly.
- In general, the harmonic mean is used when there is a necessity to give greater weight to the smaller items.
- It is applied in the case of times and average rates.

For a set of observations $x_1, x_2, \dots, x_n$; ($x_i > 0, \forall i = 1, 2, \dots, n$), the following is the formula for the Harmonic mean,

$$HM = \left\{ \frac{1}{n} \sum_{i=1}^n \frac{1}{x_i} \right\}^{-1} = \frac{1}{\frac{1}{n} \sum_{i=1}^n \frac{1}{x_i}} = \frac{n}{\sum_{i=1}^n \frac{1}{x_i}}$$

Example: The Harmonic Mean

- A boat travels upstream from B to A at a speed of 15 km/hour and down stream from A to B at a speed of 25 km/hour. What is the average speed of the boat?

$$\begin{aligned} HM &= \left\{ \frac{1}{n} \sum_{i=1}^n \frac{1}{x_i} \right\}^{-1} \\ &= \left\{ \frac{1}{2} \left(\frac{1}{15} + \frac{1}{25} \right) \right\}^{-1} \\ &= 18.75 \text{ km/hour} \end{aligned}$$

Test Yourself

1. Evan was driving up a mountainous road, with some flat stretches among the steep climbs. As a result, his speed varied for each 10 kilometers, as seen below

Distance Travelled (km)	Speed (km/h)
10	10
10	25
10	6
10	15
10	8

What was his average speed throughout the drive?

2. The harmonic mean of three numbers is 10, and the first numbers is 15. The average of the first two numbers is 12. Find the other number.

Harmonic Mean for Grouped Data

- Notational form of a frequency distribution for grouped data:

Class Interval	Class mid value	Frequency
$L_1 - U_1$	x_1	f_1
$L_2 - U_2$	x_2	f_2
$L_3 - U_3$	x_3	f_3
$\dots$	$\dots$	$\dots$
$L_i - U_i$	x_i	f_i
$\dots$	$\dots$	$\dots$

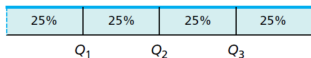
- For a frequency distribution with f_i and x_i denoting the class frequency and class mid value of the i th class ($i = 1, 2, \dots, k$), the harmonic mean can be obtained as:

$$HM = \frac{N}{\sum_{i=1}^n \frac{f_i}{x_i}}$$

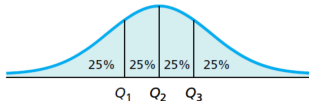
Where, $N = \sum_{i=1}^k f_i$

Quartiles, Percentiles, Deciles

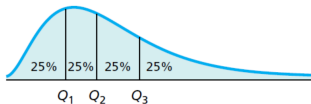
Quartiles



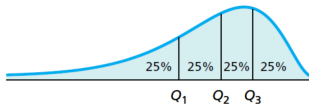
(a) Uniform



(b) Bell shaped



(c) Right skewed



(d) Left skewed

- **The first quartile (Q_1)**, is the number that divides the bottom 25% of the data from the top 75%;
- **The second quartile (Q_2)**, is the **median**, which, as you know, is the number that divides the bottom 50% of the data from the top 50%; and
- **The third quartile (Q_3)**, is the number that divides the bottom 75% of the data from the top 25%.

Quartiles

- The first quartile (Q_1) is the median of the bottom half of the data set,
- the second quartile (Q_2) is the median of the entire data set and
- the third quartile (Q_3) is the median of the top half of the data set.

To Determine the Quartiles

- Step 1: Arrange the data in increasing order.
- Step 2: Find the median of the entire data set. This value is the second quartile, Q_2 .
- Step 3: Divide the ordered data set into two halves, a bottom half and a top half.

Quartiles

To Determine the Quartiles

- Step 4: Find the median of the bottom half of the data set. This value is the first quartile, Q_1 .
 - Step 5: Find the median of the top half of the data set. This value is the third quartile, Q_3 .
 - Step 6: Summarize the results.
- A sample of 20 people yielded the weekly viewing times (in hours) displayed in the table. Determine and interpret the quartiles for these data.

25	41	27	32	43
66	35	31	15	5
34	26	32	38	16
30	38	30	20	21

Quartiles: Example 1 (n is even)

- **Step 1:** Arrange the data in increasing order.

5 15 16 20 21 25 26 27 30 30 31 32 32 34 35 38 38 41 43 66

- **Step 2:** Find the median of the entire data set. This value is the second quartile, Q_2 .

The number of observations is 20 and, consequently, the median is at position $(20 + 1)/2 = 10.5$, halfway between the tenth and eleventh observations in the ordered list. Thus, the median of the entire data set, $Q_2 = (30 + 31)/2 = 30.5$

- **Step 3:** Divide the ordered data set into two halves, a bottom half and a top half.

Referring to the ordered list in Step 1, we see that the bottom half and top half of the data set are as follows:

5	15	16	20	21	25	26	27	30	30										
										31	32	32	34	35	38	38	41	43	66
Bottom half										Top half									

Quartiles: Example 1 (n is even) (Cont...)

- **Step 4:** Find the median of the bottom half of the data set. This value is the first quartile, Q_1 .

Referring to Step 3, we see that the bottom half of the data set has 10 observations, so its median is at position $(10 + 1)/2 = 5.5$, halfway between the fifth and sixth observations in the ordered list. Thus, the median of this data set and hence the first quartile, $Q_1 = (21 + 25)/2 = 23$

- **Step 5:** Find the median of the top half of the data set. This value is the third quartile, Q_3 .

Referring again to Step 3, we see that the top half of the data set has 10 observations, so its median is at position $(10 + 1)/2 = 5.5$, halfway between the fifth and sixth observations in the ordered list. Thus, the median of this data set and hence the third quartile, $Q_3 = (35 + 38)/2 = 36.5$

Quartiles: Example 1 (n is even) (Cont...)

- **Step 6: Summarize the results.**

In summary, the three quartiles for the TV-viewing times in Table are $Q_1 = 23$ hours, $Q_2 = 30.5$ hours, and $Q_3 = 36.5$ hours.

- **Interpretation:** We see that 25% of the TV-viewing times are less than 23 hours, 25% are between 23 hours and 30.5 hours, 25% are between 30.5 hours and 36.5 hours, and 25% are greater than 36.5 hours.

Quartiles: Example 2 (n is odd)

- From Tropical Cyclone Reports, published by the National Hurricane Center, we obtained the data shown in the table on maximum wind speeds, in miles per hour (mph), for one years tropical cyclones in the Atlantic Basin. Determine and interpret the quartiles for these data.

60	70	85	65	100
60	110	45	80	40
105	80	115	90	50
45	90	115	50	

- Step 1:** Arrange the data in increasing order.

40 45 45 50 50 60 60 65 70 80 80 85 90 90 100 105 110 115 115

Quartiles: Example 1 (n is odd)

- **Step 2:** Find the median of the entire data set. This value is the second quartile, Q_2 .

The number of observations is 19 and, consequently, the median is at position $(19 + 1)/2 = 10$, the tenth observation in the ordered list. Thus, the median of the entire data set, $Q_2 = 80$.

- **Step 3:** Divide the ordered data set into two halves, a bottom half and a top half.

Because the number of observations is 19, which is an odd number, we exclude the median in both the bottom and top halves of the data set.



Quartiles: Example 1 (n is odd) (Cont...)

- **Step 4:** Find the median of the bottom half of the data set. This value is the first quartile, Q_1 .

Referring to Step 3, we see that the bottom half of the data set has 9 observations, so its median is at position $(9 + 1)/2 = 5$, the fifth observation in the ordered list. Thus, the median of this data set and hence the first quartile, $Q_1 = 50$.

- **Step 5:** Find the median of the top half of the data set. This value is the third quartile, Q_3 .

Referring again to Step 3, we see that the top half of the data set has 9 observations, so its median is at position $(9 + 1)/2 = 5$, the fifth observation in the ordered list. Thus, the median of this data set and hence the third quartile, $Q_3 = 100$.

Quartiles: Example 2 (n is odd) (Cont...)

- **Step 6: Summarize the results.**

In summary, the three quartiles for the maximum wind speeds in the table are $Q_1 = 50$ mph, $Q_2 = 80$ mph, and $Q_3 = 100$ mph.

- **Interpretation:** We see that 25% of the maximum wind speeds are less than 50 mph, 25% are between 50 mph and 80 mph, 25% are between 80 mph and 100 mph, and 25% are greater than 100 mph.

Percentile & Decile

- p th percentile is the number that divides the bottom $p\%$ of the data from the top $(100 - p)\%$.
- We use the letter P , subscripted with the percent in question, to denote a percentile.
- For example, the 30th percentile is denoted P_{30} and is the number that divides the bottom 30% of the data from the top 70%; the 2.5th percentile is denoted $P_{2.5}$ and is the number that divides the bottom 2.5% of the data from the top 97.5%. Note that the median is also the 50th percentile.
- Certain percentiles are particularly important: the 10th, 20th,..., 90th percentiles are called the deciles and divide a data set into tenths (10 equal parts).
- Note: $\text{Median} = Q_2 = D_5 = P_{50}$

Formula for Quartiles

Quartile

- For raw data,
 - if $i = 1, 2, 3$
 - $n =$ number of values
- when $\frac{i \times n}{4}$ is an integer

$$i^{th} \text{ quartile}, Q_i = \frac{1}{2} \left[\left(\frac{i \times n}{4} \right)^{th} \text{ value} + \left(\frac{i \times n}{4} + 1 \right)^{th} \text{ value} \right],$$

- when $\frac{i \times n}{4}$ is not an integer

$$i^{th} \text{ quartile}, Q_i = \text{next integer value of } \frac{i \times n}{4},$$

Formula for Deciles

Decile

- For raw data,
 - if $i = 1, 2, 3, \dots, 9$
 - $n = \text{number of values}$
- when $\frac{i \times n}{10}$ is an integer

$$i^{\text{th}} \text{ Decile}, D_i = \frac{1}{2} \left[\left(\frac{i \times n}{10} \right)^{\text{th}} \text{ value} + \left(\frac{i \times n}{10} + 1 \right)^{\text{th}} \text{ value} \right],$$

- when $\frac{i \times n}{10}$ is not an integer

$$i^{\text{th}} \text{ Decile}, D_i = \text{next integer value of } \frac{i \times n}{10},$$

Formula for Percentiles

Percentile

- For raw data,
 - if $i = 1, 2, 3, \dots, 99$
 - $n =$ number of values
- when $\frac{i \times n}{100}$ is an integer

$$i^{th} \text{ percentile}, P_i = \frac{1}{2} \left[\left(\frac{i \times n}{100} \right)^{th} \text{ value} + \left(\frac{i \times n}{100} + 1 \right)^{th} \text{ value} \right],$$

- when $\frac{i \times n}{100}$ is not an integer

$$i^{th} \text{ percentile}, P_i = \text{next integer value of } \frac{i \times n}{100},$$

Example

- Following is the marks in STA 201 obtained by 20 students in Spring 2024.

99	75	84	33	45	66	97	69	55	61
72	91	74	93	54	76	62	91	77	68

- Find out
 - 1^{st} quartile & 3^{rd} quartile
 - 3^{rd} decile, 6^{th} decile
 - 20^{th} percentile, 37^{th} percentile & 86^{th} percentile

Solution

- First, arrange the data

33	45	54	55	61	62	66	68	69	72
74	75	76	77	84	91	91	93	97	99

- Here, $n = 20$
- For 1st quartile : $i = 1, \frac{i \times n}{4} = \frac{1 \times 20}{4} = 5$, an integer.
- So,

$$\begin{aligned} 1^{\text{st}} \text{ quartile} &= \frac{1}{2} [5^{\text{th}} \text{ value} + (5 + 1)^{\text{th}} \text{ value}] \\ &= \frac{1}{2} (61 + 62) = 61.5 \end{aligned}$$

Solution Cont...

- 6^{th} Decile : for $i = 6$, $\frac{i \times n}{10} = \frac{6 \times 20}{10} = 12$, an integer
- so,

$$\begin{aligned} 6^{th} \text{ Decile, } D_6 &= \frac{1}{2} [12^{th} \text{ value} + (12 + 1)^{th} \text{ value}] \\ &= \frac{1}{2}(75 + 76) = 75.5 \end{aligned}$$

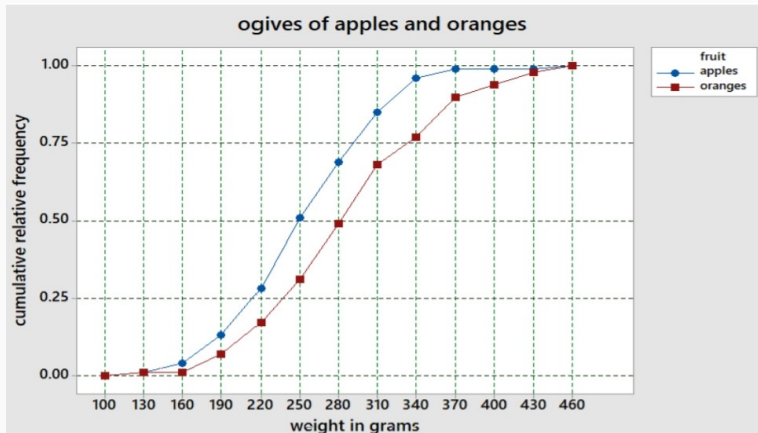
- 37^{th} percentile : for $i = 37$, $\frac{i \times n}{100} = \frac{37 \times 20}{100} = 7.4$
- not an integer. So, next integer = 8.

$$37^{th} \text{ percentile, } P_{37} = 8^{th} \text{ value} = 68$$

Calculate Quartile, Percentile & Decile from Ogive

Find the 25th, 50th and 75th percentile for apple and orange from the following figure:

Follow the class lecture



Thank You